Reading sea-wave dynamics from video

A multi-scale toolbox for cross-modal coupling

`HNA.modalities.video`

HumanNatureAttunement project
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Abstract

A wave is at once a regular oscillator and an unpredictable turbulent surface, and human physiology – breath, heartbeat, brain rhythms – responds to both faces of it. To make that responsiveness measurable the HumanNatureAttunement project needs the visual modality on the same footing as the audio side: a battery of 1-D signals, sampled at the frame rate, that capture what the eye is doing without losing multi-scale structure. `HNA.modalities.video` delivers exactly that, organised along two axes – a *spatial scale* of analysis (whole-image, per-patch, per-pixel) and a *feature family* (**raw**, **oscillatory**, **complexity**). The product is a 3-by-3 framework, nine cells of extractors whose 1-D outputs are drop-in inputs to the project's coupling toolbox. We illustrate every cell with a single set of four contrasting clips and close with a cross-clip fingerprint that visibly groups by family.

The Python sub-package `src/HNA/modalities/video/` is a one-to-one mirror of the framework: each cell of Fig. 1 maps to a file in that directory.

Contents

1 What this module is for

The HumanNatureAttunement project asks whether the human body and brain *entrain* to natural rhythmic stimuli, and whether the strength of that entrainment depends on attention or on the listener's state. The audio-side pipeline is established: the acoustic envelope of a wave recording is correlated with EEG band power, and the coupling is stronger under undivided attention. The video module extends the question symmetrically: *does the brain (or the heart, or the muscles) track the visual envelope of the same stimulus?*

That symmetry has a cost: a wave looks like many things at once. A breaking crest is a sharp **edge**; the swell is a slow horizontal **stripe** travelling across the frame; foam break-up is a turbulent **fractal texture**. Each has its own characteristic timescale and its own plausible coupling target on the physiological side. The toolbox therefore gives us a generous menu of 1-D signals, each describing a different visual facet of the same clip.

Design rules

Every signal is **1-D, sampled at the frame rate**, **physically interpretable**, and **cheap**: minutes of HD footage in seconds on CPU. These three rules let the same coupling tools that already work on EEG, HRV and audio plug directly into the visual modality.

2 The framework, in one figure

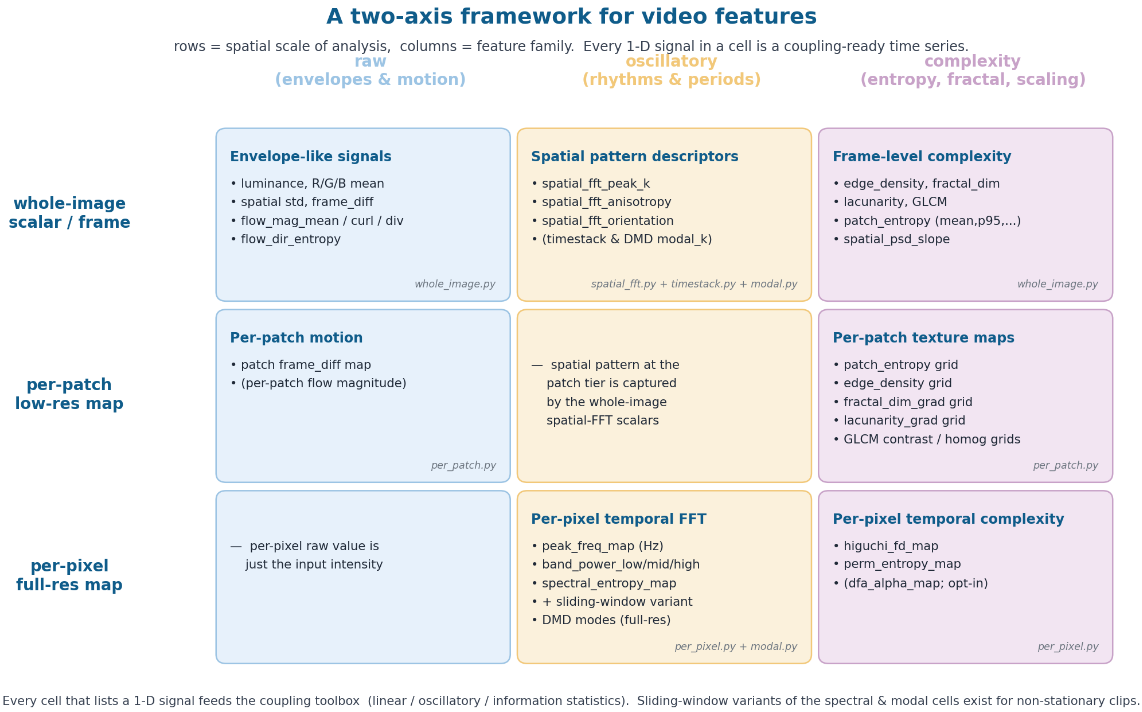

Figure 1: **Two axes.** *Rows* are the spatial scale at which a measurement is made: one scalar per frame (top), a low-resolution patch grid (middle), or one number per pixel (bottom). *Columns* are the family of question we ask of the scene: a **raw value**, an **oscillation rate**, or a **complexity summary**. Annotations under each cell name the source file in `src/HNA/modalities/video/`, so the figure doubles as a code map.

Spatial scale (rows).

- *Whole-image* – one scalar per frame.
- *Per-patch* – a coarse $\sim 8 \times 8$ grid of tile scalars per frame, returned as both a low-resolution map (visualisation) and a per-frame spatial mean (coupling).
- *Per-pixel* – every pixel of a downscaled frame keeps its own time series; the output is one full-resolution 2-D map per quantity.

Feature family (columns).

- **Raw** – instantaneous values: luminance, colour means, frame difference, optical-flow magnitude / curl / divergence. The natural pair for a physiological raw envelope.
- **Oscillatory** – "what frequency?". Timestack peak, DMD modes, 2-D-FFT spatial wavenumber, per-pixel temporal peak. The natural pair for a physiological rhythm (HRV LF/HF peak, EEG alpha peak).
- **Complexity** – "how irregular?". Edge density, fractal dimension, lacunarity, GLCM, patch entropy, $1/f$ slope. The natural pair for a complexity time series on the physiology side.

The empty corner (per-pixel **raw**) is just the input intensity itself and not interesting as a feature. Every other cell is populated.

3 Four sea states as running examples

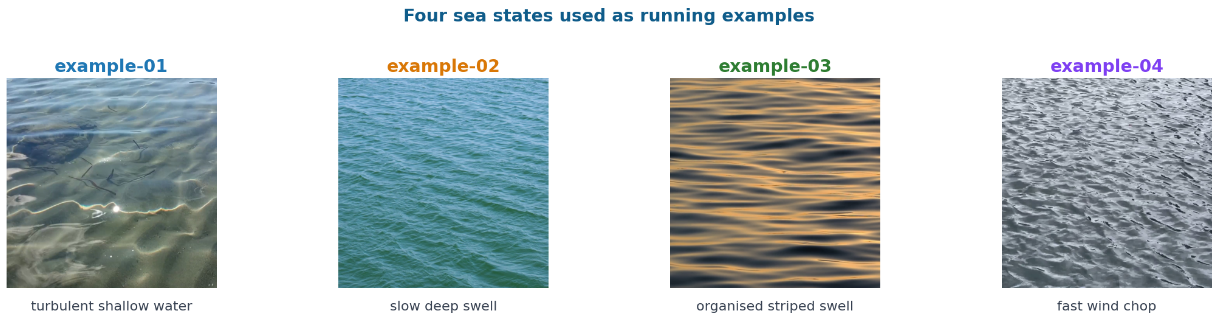

Figure 2: The four example clips used throughout this report. They bracket realistic conditions: a turbulent shallow- water surf zone; a slow deep swell; an organised striped swell dominated by horizontal crests; and fast wind-driven chop. Every figure below uses the same four clips and the same colour key.

We picked four physically distinct clips that we expect to score differently on most of the toolbox's metrics (Fig. 2). ex-01 has the most motion and the most local foam structure; ex-02 is the slowest and the most laminar; ex-03 is the most spatially regular (dominant horizontal stripes); ex-04 sits at the high-frequency end (fast wind chop, fine-scale texture). Re-using the same four clips in every figure keeps the reader's eye calibrated.

4 Raw envelopes & motion

The cheapest, most directly comparable family. A grayscale frame collapses to seven scalars: mean luminance, per-channel R/G/B means, spatial standard deviation, a 64-bin Shannon entropy of intensity, and the frame-to-frame absolute difference. Adding dense optical flow gives mean magnitude, a 95th-percentile "surge" detector, mean |curl| (rotational / turbulent content), mean divergence, and the entropy of flow direction.

Fig. 3 aligns three of these traces – luminance, flow magnitude, frame-difference – across the four clips on a shared z-score axis. The four sea states differ visibly in motion energy and rhythm regularity: the swell of ex-02 dominates its luminance trace; ex-01 oscillates broadly; ex-03's slow drift is striking against its high-frequency frame-difference; ex-04 packs every luminance value into a tight, uniform high-frequency oscillation.

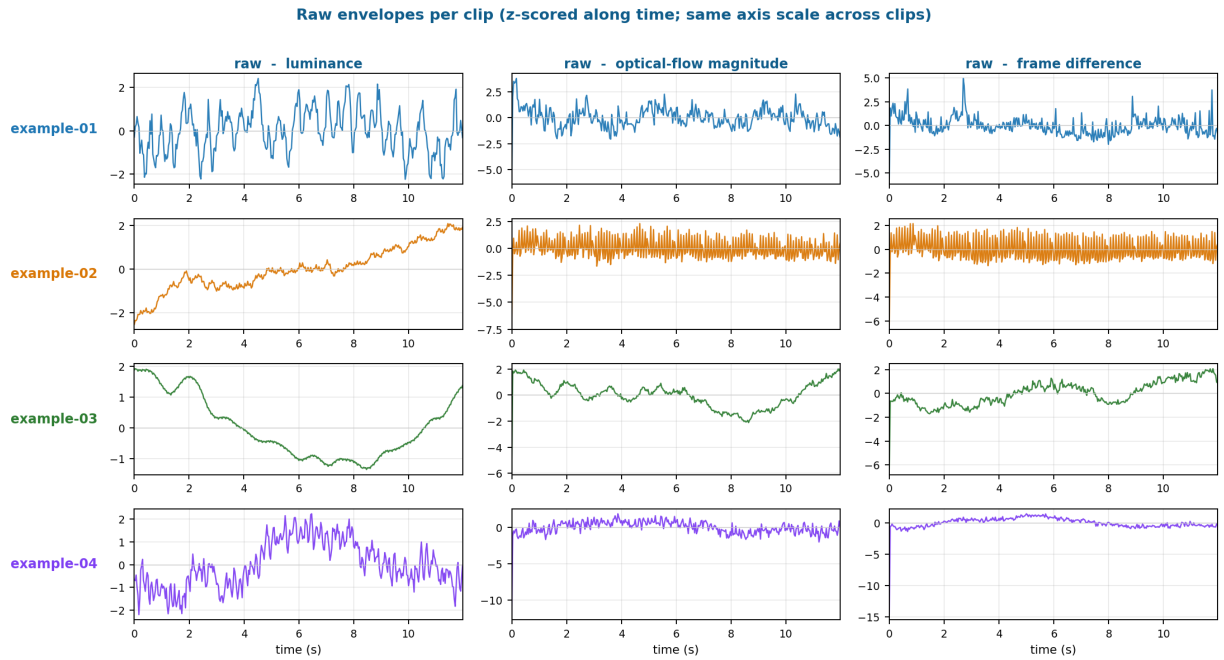

Figure 3: Three **raw** envelopes per clip (z-scored along time, same axis scale). Luminance, mean optical-flow magnitude, and per-frame absolute difference. These are the most direct visual analogue of an audio envelope; they pair naturally with respiration or EEG band-power on the coupling side.

Multi-scale flow. Optical-flow magnitude at three *temporal strides* $\Delta t \in \{1, 3, 9\}$ frames isolates ripple-scale motion ($\Delta t = 1$), intermediate motion, and quarter-swell motion ($\Delta t = 9$ at ~ 30 fps probes ~ 0.3 s). The same four scalars (mag, |curl|, div, direction entropy) are then computed on the full field, on its Gaussian low-pass component (swell), and on the residual (turbulence), giving 36 frame-rate signals from one extractor (Fig. 4).

5 Oscillatory: rhythms and dominant periods

Three extractors live in this column at three different spatial scales. Together they answer "*what is this clip's heartbeat?*".

5.1 Timestack – whole-image scalar

Borrowed from coastal remote sensing (**?**): sample one vertical column at every frame, average its rows, take the Welch PSD, report the band-restricted argmax. That gives one scalar per clip (the dominant wave frequency) plus the 1-D row-mean signal as a coupling input. We measure 1.50 Hz for ex-01 and 0.12 Hz for the other three – consistent with a fast-chopping shallow-water surf zone vs. slower deep-water swells.

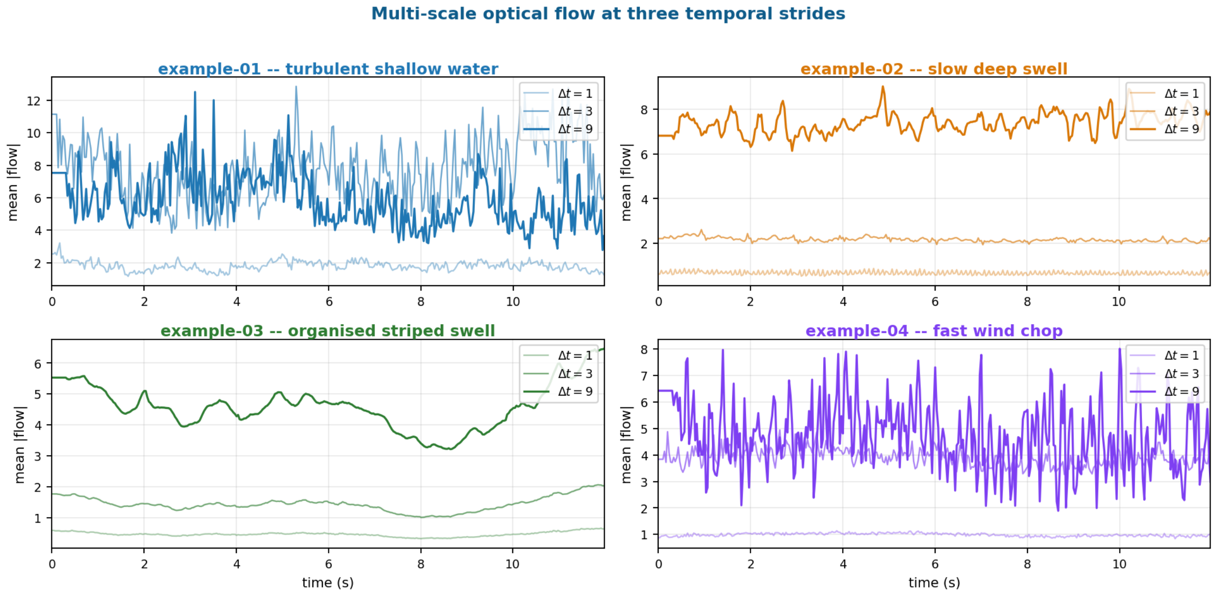

Figure 4: Multi-scale optical flow at the three strides, per clip. The longer the baseline, the more swell motion is integrated – hence the systematic offset between $\Delta t = 1$ and $\Delta t = 9$ in every panel. ex-02 (slow deep swell) shows the largest stride-9 mean: a slow but spatially coherent drift moves a lot of pixels per nine-frame jump.

5.2 Modal decomposition (DMD) – per-pixel-resolution modes

DMD (**???**) fits one linear evolution operator $X_{t+1} = AX_t$ to the (pixels × time) matrix and returns top-k *complex* eigenvalues, so each spatio-temporal mode has a single frequency. (An SVD/POD baseline pairs sin/cos at the same frequency: a single-component sea at $f_p = 0.125$ Hz produces four POD modes at $\{0.125, 0.125, 0.250, 0.250\}$ Hz – visually similar and spectrally indistinguishable. DMD removes that redundancy.)

The frequencies ladder up cleanly across the four clips: ex-03 places its top modes near 0.18 – 0.29 Hz (a slow regular swell); ex-02 puts both top modes around 1.0 Hz; ex-01 has two distinct modes at 0 Hz (a stationary mean-removed component) and 0.12 Hz; ex-04 sits at 2.5 Hz and above (Fig. 5). Each mode's real-valued temporal coefficient is exposed as a 1-D signal in `feats.signals` under the keys `modal_1`, `modal_2`, ...

5.3 Per-pixel temporal FFT – per-pixel maps

The most spatially detailed of the three: every pixel of a downscaled copy keeps its own dominant frequency, its own spectral entropy, its own band powers. The result is a small set of 2-D maps and two clip-level scalars – `sync_index` (concentration of the mean spectrum) and `coherence_index` (spatial uniformity of peak frequency). The maps themselves answer "*which part of the frame is the swell, which part is the foam?*" (Fig. 6).

5.4 Spatial pattern (2-D FFT)

A different kind of "oscillatory": not in time but in space. The 2-D Fourier transform of a single frame gives a peak radial wavenumber (the dominant spatial period of foam / ripple texture), a horizontal-vs-vertical anisotropy (positive when horizontal swell crests dominate), and a weighted-mean orientation. In our four-clip fingerprint (Fig. 8, oscillatory band) the anisotropy peaks at 3.11 on ex-03 – the only clip with a tightly aligned striped swell – and is positive on every clip we sampled.

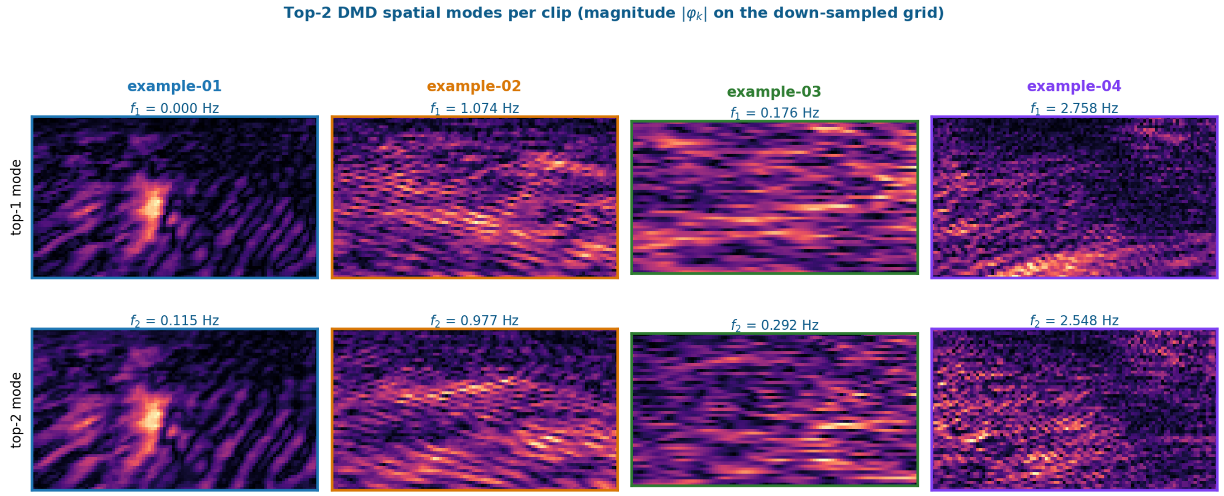

Figure 5: Top-2 **DMD spatial modes** for each clip, $|\varphi_k|$ on the down-sampled grid. Frequencies $f_k = |\omega_k|/2\pi$ printed above each panel. **The modes are derived per clip**: each clip has its own decomposition, and the spatial patterns differ accordingly – ex-01 concentrates energy in foreground foam, ex-02 shows a smooth large-scale swell pattern, ex-03 resolves the dominant horizontal stripe of the regular swell, ex-04 is fine-scale turbulent texture at high frequency.

5.5 Sliding-window variants for non-stationary clips

Three of the cells above – per-pixel FFT, DMD, and timestack PSD – are by default *single-window* estimators: they fit one model to the whole clip. Whenever the clip is long enough that the dominant frequency may drift (a swell building, the wind freshening), a single window is too coarse. We expose sliding-window counterparts that promote those summaries to 1-D signals at frame rate: `extract_pixel_spectrum_windowed`, `extract_modal_windowed`, `extract_timestack_windowed`. They share the parameters `windowed_win_sec` (default 4 s) and `windowed_step_sec` (default 0.5 s); per-window values are linearly interpolated onto the frame grid so they line up with everything else.

6 Complexity: roughness, clumpiness, irregularity

The third column collects every measure that asks "*how complex is this?*" rather than "*what value, what frequency?*".

Frame-level scalars. Edge density (Canny), the radial-PSD slope (parallel to the EEG aperiodic $1/f$ exponent), box-counting fractal dimensions on two binarisations of the frame (Canny edges and gradient-magnitude), Plotnick / Allgood lacunarity on those same binarisations, GLCM contrast / homogeneity, and the patch-Shannon-entropy mean. D (fractal dimension) and Λ (lacunarity) are *complementary*: D is nearly identical across the four sea states (they all sit near 2.05), but Λ on the gradient image splits them strongly (Fig. 7).

Per-patch maps (`extract_spatial_field`). The same frame-level descriptors recomputed inside every patch of an $\sim 8\times 8$ grid. Each call returns both a (G_y, G_x, T) stack of patch maps for visualisation and the per-frame spatial mean of each map as a 1-D signal (suffix `field_*`). This fills the per-patch row of the framework – previously sparse – and surfaces the within-frame heterogeneity that the clip-level mean throws away.

Per-pixel temporal complexity (`extract_pixel_complexity`). The deepest spatial reduction: at every pixel of a downscaled copy, run a Higuchi fractal dimension and a permutation entropy (and optionally DFA) on the luminance time series. The output is a full-resolution 2-D map per

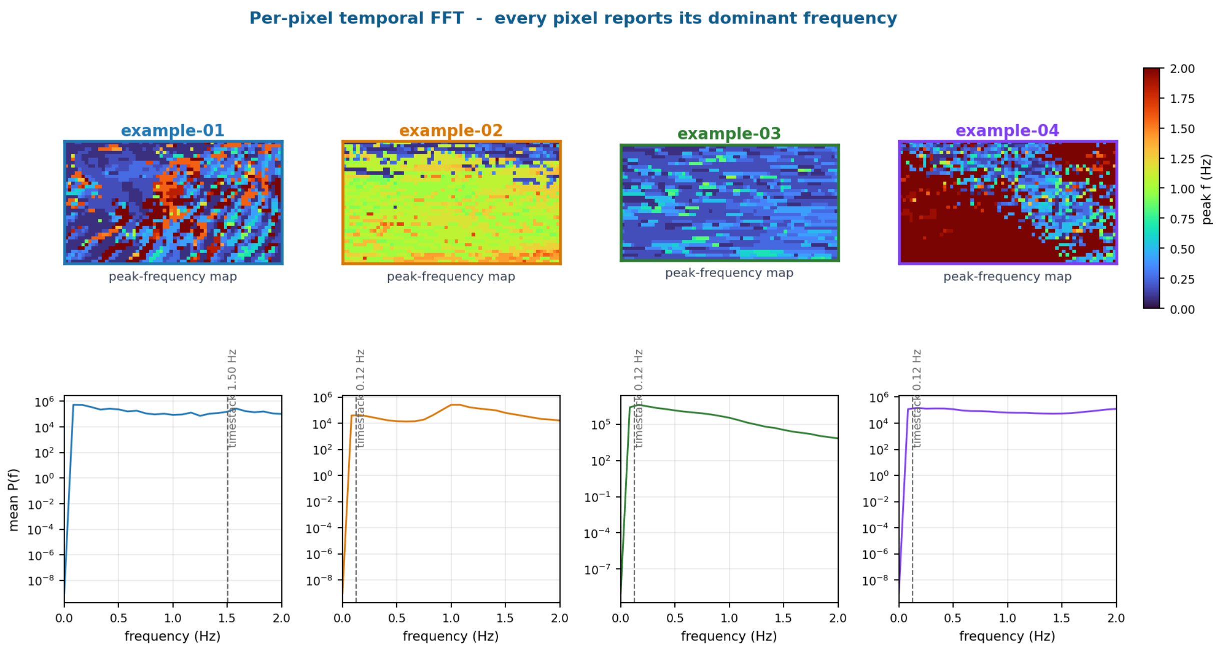

Figure 6: Per-pixel temporal FFT. Top: peak-frequency map – every pixel coloured by the frequency at which it oscillates the most. Bottom: mean spectrum across pixels, with the timestack peak marked for reference. Note the clean spatial structure on ex-02 (a near-uniform $\sim$1 Hz pulse over the whole frame) versus the fragmented, foam-dominated map on ex-01; ex-04 divides into high-frequency wind-chop (red) and slow background (blue).

measure – "*where on the frame is the temporal behaviour most complex?*". Opt-in because it is the toolbox's most expensive operation.

The 15-measure summary, applied to every 1-D signal. Independently of the maps, every 1-D signal in `feats.signals` is also summarised by 15 nonlinear-dynamics measures via `complexity_summary`: five entropies, three fractal dimensions, Hjorth mobility / complexity, DFA, Hurst, Lempel–Ziv, multiscale entropy area, and the $1/f$ slope. The reason to repeat them on every signal is that "complexity" is multifaceted: two signals can match on entropy and disagree on fractal dimension. Reporting all 15 and letting the downstream analysis pick keeps options open. The same library implementations (`antropy`, `neurokit2`) are used on the EEG/HRV side, so cross-modal complexity comparisons are apples-to-apples.

7 Cross-clip fingerprint, organised by family

The reason this figure exists is exactly the point we keep coming back to: *no single video metric is a perfect summary, and that is the point.* Each metric is sensitive to a different physical facet of the scene; the union is the clip's signature. Reading horizontally tells you which clip wins on a given metric; reading vertically tells you the clip's identity. The family-grouped layout makes the second reading much easier than a flat heatmap because the question is no longer "*is this row important?*" but "*which family is most informative for this clip?*".

8 From features to coupling

Every signal in `feats.signals` is a 1-D `np.ndarray` sampled at f_s = `feats.fs` and is therefore an immediate input to `HNA.modules.coupling`. The coupling toolbox itself organises into three families that mirror the feature columns (Table 1).

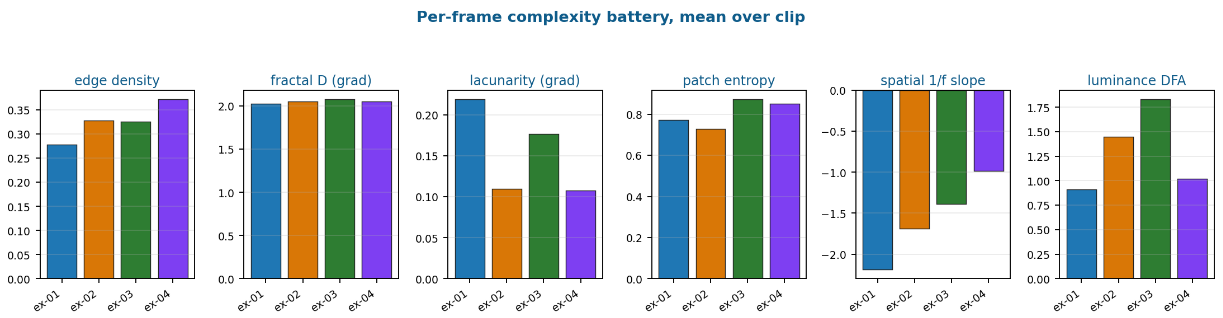

Figure 7: Six **complexity** measures, mean over each clip. Edge density and patch entropy push ex-04 highest (foam everywhere, broad intensity histogram in every patch); fractal dimension is essentially flat across clips, but lacunarity is not; and the spatial $1/f$ slope is steepest for ex-01 (heavy low-spatial-frequency content from the foreground swell). Luminance DFA, by contrast, is highest for ex-03, where the slow regular swell makes the luminance envelope look long-range correlated.

Coupling family	Statistics	Pair naturally with
Linear	windowed cross-correlation, magnitude-squared coherence, peak r + lag	**raw** envelopes (`luminance`, `flow_mag_mean`, `frame_diff`) vs. respiration / EEG band power.
Oscillatory	phase-locking value (PLV), weighted phase-lag index (wPLI)	**oscillatory** signals (`modal_1`, `timestack_w_peak_freq_hz`, `pixspec_w_peak_freq_median`) vs. HRV LF/HF peak, EEG alpha-peak frequency.
Information	mutual information, transfer entropy	**complexity**-paired signals (`wc_flow_mag_mean__higuchi_fd`, `wc_patch_entropy__perm_entropy`) vs. rolling Higuchi FD or permutation entropy on EEG / HRV.

Table 1: Three coupling families on the physiology side, mirrored to the three feature families on the video side. The right column shows pairings where “same family on both sides” yields the most interpretable estimate.

```python
from HNA.modalities.video import quantify_video
from HNA.modules.coupling import (
    windowed_xcorr, band_coherence,        # linear
    plv_phase_sync, wpli_phase_sync,        # oscillatory
)

feats = quantify_video("video_examples/example-01.mp4",
                       include_pixel_spectrum_windowed=True,
                       include_modal_windowed=True,
                       include_timestack_windowed=True,
                       include_spatial_field=True,
                       include_pixel_complexity=True)

# linear: video luminance vs. respiration envelope
xc  = windowed_xcorr(feats.signals["luminance"], respiration,
                     fs=feats.fs, win_sec=180, step_sec=10,
                     max_lag_sec=30)
# oscillatory: instantaneous wave frequency vs. EEG alpha-peak
```

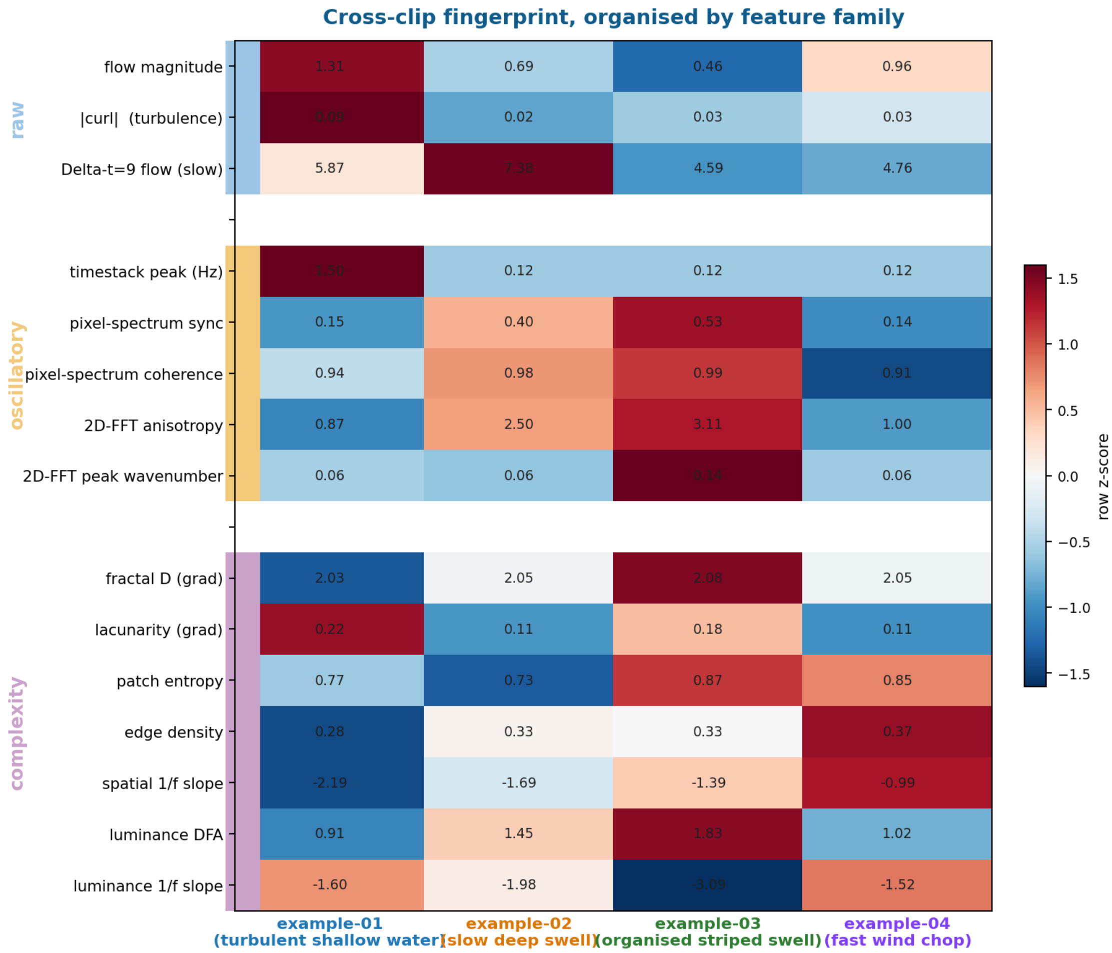

Figure 8: Cross-clip fingerprint with rows grouped by feature family (left side stripes). Each cell text is the raw metric; the colour is the row z-score across the four clips, so within each row the most extreme cell is the most informative. The blocks make a visible structural point: a clip's signature is not a single number but a *pattern of agreements and disagreements* across families. ex-01 is the only clip where the **timestack peak** is dominant; ex-03 owns almost every **oscillatory** cell (high pixel-spectrum sync, high coherence, high 2-D-FFT anisotropy); ex-04 wins the **complexity** bands on edge density and gradient fractal dimension.

```
plv = plv_phase_sync(feats.signals["timestack_w_peak_freq_hz"],
                     eeg_alpha_peak_f, fs=feats.fs)
```

9 Practical recipe

Default call.

```
from HNA.modalities.video import quantify_video
feats = quantify_video("path/to/clip.mp4")
```

That gets you envelopes, single-scale optical flow, frame-level spatial complexity, 2-D-FFT scalars, the timestack, top-4 DMD modes, and the 15-measure complexity summary on every signal. About 5 seconds for a 20-second HD clip on CPU.

Add what you need. The opt-in flags map one-to-one to cells of Fig. 1:

- `include_spatial_field=True` – per-patch maps + means (per-patch row, all three feature columns).

- `include_pixel_spectrum=True` – per-pixel temporal FFT maps (per-pixel **oscillatory**).
- `include_pixel_complexity=True` – per-pixel temporal complexity maps (per-pixel **complexity**; opt-in because it is the most expensive operation).
- `include_pixel_spectrum_windowed=True` / `include_modal_windowed=True` / `include_timestack_windowe` – the sliding-window counterparts, for non-stationary clips.
- `include_windowed_complexity=True` – promote any 1-D signal's complexity to a 1-D series.

File layout under `src/HNA/modalities/video/`. The sub-package mirrors Fig. 1 cell by cell:

```
src/HNA/modalities/video/
  _common.py             # frame I/O, probe, iter_frames, _stack_video
  whole_image.py         # global signals, optical flow, spatial complexity
  spatial_fft.py         # per-frame 2-D-FFT scalars
  per_patch.py           # extract_spatial_field
  per_pixel.py           # pixel_spectrum + extract_pixel_complexity
  modal.py               # DMD (global + windowed)
  timestack.py           # timestack (global + windowed)
  temporal_complexity.py # complexity_summary, windowed_complexity
  pipeline.py            # quantify_video
```

The public API is exposed at the top level of the sub-package: `from HNA.modalities.video import quantify_video, extract_<name>, ...`.

Defaults worth knowing. The whole pipeline runs on a 192-px-on-the-long-axis grayscale copy of the video; the modal decomposition uses a 96-px copy; per-pixel complexity uses an even smaller 48-px copy by default. Patch size is 24 px ($\sim 8\times 8$ grid). Sliding-window defaults are $W = 4$ s, $h = 0.5$ s for the spectral / modal / timestack family and $W = 2$ s, $h = 0.25$ s for the complexity family. The full hyperparameter list is documented inline in each extractor's docstring.

Reproducibility

```
# 1. Install the package with the video extras. opencv-python is
#    required to decode video; pydmd is optional (the modal extractor
#    falls back to a POD/SVD baseline if missing). scipy / antropy /
#    neurokit2 are base deps of the toolbox and already get installed
#    as part of the regular package.
pip install -e .[video]

# 2. Quantify a single clip
python -c "from␣HNA.modalities.video␣import␣quantify_video;␣\
␣␣␣␣␣␣␣␣␣␣␣␣f␣=␣quantify_video('video_examples/example-01.mp4');␣\
␣␣␣␣␣␣␣␣␣␣␣␣print(len(f.signals),␣'signals␣at',␣f.fs,␣'Hz')"

# 3. Re-render the report figures
python scripts/figures/video_v3/framework.py   # framework.png
python scripts/figures/video_v3/all_figures.py         # v3_*.png

# 4. Compile the report
tectonic reports/video_v3/video_v3.tex
```